From the Weight-Space Cyclic Locus to the Critical Locus

Abstract

We revisit the geometry of rank-r critical loci of deep linear networks from a direct weight-space viewpoint. In previous versions of the project, the passage from the cyclic variety to the critical locus was mediated by a residual parameter space together with a lifting map. Here we keep both objects directly inside the parameter space Ω. We define the cyclic locus by the transpose-gradient equations

$$W_{<\ell} W_0 W_{>\ell} = 0, \qquad W_0 = \Sigma_{XY} P_Q,$$

and study its rank-r part $\Gamma^r_{d,S}$. The critical locus $\mathcal{C}^r_{d,S}$ is then the distinguished fiber cut out by the additional condition $\mu(W) = P_S W^*$. The main result of this draft is that the $G_d^{W_0}$-orbits in $\Gamma^r_{d,S}$ are in bijection with the H_S-orbits in $\mathcal{C}^r_{d,S}$. This gives the direct weight-space replacement for the old lifting statement and shows formally how a stratification of the cyclic locus transfers to the critical locus.

1 Introduction

The motivation of this draft is to keep the geometry of critical points inside the original parameter space of a deep linear network. In the residual-space approach of the earlier versions, one first encoded the cyclic equations in a smaller residual variety and then lifted the corresponding data back to the critical locus by a block-normal-form construction. That point of view is effective, but it hides the geometric relation between the cyclic equations and the actual weight matrices.

The present draft works directly in weight space. The cyclic equations are written as polynomial equations in the weights, the determinantal and critical loci are defined directly in Ω, and the passage from the cyclic locus to the critical locus is expressed as an orbit correspondence rather than as a lifting construction. This shift in viewpoint is the main structural difference with `paper_v1.tex`, and it is also the main motivation for the paper.

2 Setup and Background

2.1 Notation

Notation	Meaning
L	number of weight matrices in the deep linear network
$d = (d_0, \dots, d_L)$	dimension vector
Ω	parameter space $\prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R})$
μ	product map $\mu(W) = W_L \cdots W_1$
W^*	teacher matrix $W^* = \Sigma_{YX}\Sigma_X^{-1}$
$r_{\max}$	rank of W^*
$S \subset \{1, \dots, r_{\max}\}$	subset of singular directions, with $\lvert S\rvert = r$
P_S, P_Q	complementary orthogonal projectors on $\mathbb{R}^{d_L}$
$W_0 = W_S$	closing map $\Sigma_{XY}P_Q$
Σ_d^r	rank-r determinantal locus in weight space
$\Gamma_{d,S}$	weight-space cyclic locus
$\Gamma_{d,S}^r$	rank-r cyclic locus
$\mathcal{C}_{d,S}^r$	rank-r critical locus attached to S
G_d	full base-change group $\prod_{\ell=0}^{L} \mathrm{GL}(d_\ell, \mathbb{R})$
G_{hid}	hidden-layer gauge group $\prod_{\ell=1}^{L-1} \mathrm{GL}(d_\ell, \mathbb{R})$
$G_d^{W_0}$	stabilizer of the closing map W_0 in G_d
H_S	stabilizer in $G_d^{W_0}$ of the distinguished product $P_S W^*$

2.2 Deep linear networks

Definition 2.1 (Deep linear network and product map)**.** Let

$$\Omega := \prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R}), \qquad W = (W_1, \dots, W_L) \in \Omega.$$

The associated deep linear network is the map

$$x \longmapsto W_L \cdots W_1 x,$$

and its end-to-end matrix is given by the product map

$$\mu : \Omega \to \mathrm{Mat}_{d_L \times d_0}(\mathbb{R}), \qquad \mu(W) := W_L \cdots W_1.$$

For each layer ℓ we use the partial-product notation

$$W_{>\ell} := W_L \cdots W_{\ell+1} \in \mathrm{Mat}_{d_L \times d_\ell}(\mathbb{R}), \qquad W_{<\ell} := W_{\ell-1} \cdots W_1 \in \mathrm{Mat}_{d_{\ell-1} \times d_0}(\mathbb{R}),$$

with the conventions

$$W_{>L} = I_{d_L}, \qquad W_{<1} = I_{d_0}.$$

Then

$$\mu(W) = W_{>\ell} W_\ell W_{<\ell} \qquad \text{for every } 1 \le \ell \le L.$$

Definition 2.2 (Population loss and teacher matrix)**.** The population quadratic loss is

$$\mathcal{L}(W) := \mathbb{E}\big[\|Y - \mu(W)X\|_2^2\big].$$

Writing

$$\Sigma_X = \mathbb{E}[XX^\top], \qquad \Sigma_{XY} = \mathbb{E}[XY^\top], \qquad \Sigma_{YX} = \Sigma_{XY}^\top,$$

the corresponding teacher matrix is

$$W^* := \Sigma_{YX}\Sigma_X^{-1}, \qquad r_{\max} := \operatorname{rank}(W^*).$$

Standing assumptions. Throughout this draft we assume that Σ_X is invertible, that Σ_{XY} has full rank d_L, that W^* has distinct nonzero singular values, and that the network has no bottleneck, i.e.

$$d_\ell \geq d_L \qquad \text{for every } 0 \leq \ell \leq L.$$

2.3 Gradient equations in weight space

Proposition 2.3 (Gradient formula)**.** For each $\ell \in \{1, \dots, L\}$,

$$\nabla_{W_\ell}\mathcal{L}(W) = 2\, W_{>\ell}^\top\big(\mu(W)\Sigma_X - \Sigma_{YX}\big)W_{<\ell}^\top.$$

Consequently,

$$\big(\nabla_{W_\ell}\mathcal{L}(W)\big)^\top = 2\, W_{<\ell}\big(\Sigma_X\mu(W)^\top - \Sigma_{XY}\big)W_{>\ell}.$$

Proof. The differential of the product map in the direction $\dot{W}_\ell$ is

$$D\mu(W)[\dot{W}_\ell] = W_{>\ell}\dot{W}_\ell W_{<\ell}.$$

Since the gradient of the quadratic loss with respect to the end-to-end matrix M is

$$\nabla_M\mathcal{L}(M) = 2(M\Sigma_X - \Sigma_{YX}),$$

the chain rule gives the first identity. The second follows by transposition and the symmetry of Σ_X. □

The direct weight-space cyclic equations arise by evaluating the transpose gradient on the rank-r critical sheet.

Proposition 2.4 (Weight-space cyclic equations)**.** Let W be a first-order critical point of rank r, and let $S \subset \{1, \dots, r_{\max}\}$ be the unique subset of cardinality r such that

$$\mu(W) = P_S W^*.$$

Then the transpose-gradient equations are equivalent to

$$W_{<\ell}W_0W_{>\ell} = 0, \qquad 1 \leq \ell \leq L,$$

where

$$W_0 = W_S := \Sigma_{XY}P_Q.$$

Proof. At a critical point associated with S, one has

$$\mu(W) = P_S W^* = P_S \Sigma_{YX} \Sigma_X^{-1}.$$

Hence

$$\Sigma_X \mu(W)^\top - \Sigma_{XY} = \Sigma_X \Sigma_X^{-1} \Sigma_{XY} P_S - \Sigma_{XY} = \Sigma_{XY}(P_S - I) = -\Sigma_{XY} P_Q.$$

Substituting this identity into the transpose-gradient formula yields the claimed equations. □

Definition 2.5 (Determinantal, cyclic, and critical loci)**.** Fix a rank r and a subset $S \subset \{1, \dots, r_{\max}\}$ with $|S| = r$. Define:

$$\Sigma_d^r := \{W \in \Omega : \operatorname{rank}(\mu(W)) = r\},$$

$$\Gamma_{d,S} := \{W \in \Omega \mid W_{<\ell} W_0 W_{>\ell} = 0 \text{ for every } 1 \le \ell \le L\},$$

$$\Gamma_{d,S}^r := \Sigma_d^r \cap \Gamma_{d,S},$$

and

$$\mathcal{C}_{d,S}^r := \left\{ W \in \Gamma_{d,S}^r \;\middle|\; \mu(W) = P_S W^* \right\}.$$

Remark 2.6 (Direct weight-space viewpoint)**.** This is the main departure from the residual-space setup of `paper_v1.tex`. The cyclic locus and the critical locus are both now defined directly in Ω, and the problem is to relate them without passing through a separate residual parameter space.

3 From the Cyclic Variety to the Critical Locus

3.1 Symmetry groups and the corrected orbit problem

Definition 3.1 (Relevant symmetry groups)**.** The full base-change group is

$$G_d := \prod_{\ell=0}^{L} \mathrm{GL}(d_\ell, \mathbb{R}),$$

acting on Ω by

$$(g \cdot W)_\ell := g_\ell W_\ell g_{\ell-1}^{-1}.$$

Its hidden subgroup is

$$G_{\mathrm{hid}} := \prod_{\ell=1}^{L-1} \mathrm{GL}(d_\ell, \mathbb{R}),$$

viewed as the subgroup of G_d with $g_0 = I_{d_0}$ and $g_L = I_{d_L}$. Finally, the stabilizer of the closing map is

$$G_d^{W_0} := \{g \in G_d : g_0 W_0 g_L^{-1} = W_0\}.$$

Lemma 3.2 (Equivariance of the product map)**.** For every $g \in G_d$ and every $W \in \Omega$,

$$\mu(g \cdot W) = g_L \mu(W) g_0^{-1}.$$

In particular, G_{hid} preserves $\mu(W)$ exactly.

Proof. Expanding the product and cancelling adjacent inverse factors gives

$$\mu(g \cdot W) = (g_L W_L g_{L-1}^{-1}) \cdots (g_1 W_1 g_0^{-1}) = g_L W_L \cdots W_1 g_0^{-1} = g_L \mu(W) g_0^{-1}.$$

If $g \in G_{\text{hid}}$, then $g_0 = I_{d_0}$ and $g_L = I_{d_L}$, hence

$$\mu(g \cdot W) = \mu(W).$$

□

Proposition 3.3 (Stability of the cyclic locus)**.** The loci $\Gamma_{d,S}$ and $\Gamma_{d,S}^r$ are stable under the action of $G_d^{W_0}$.

Proof. For fixed ℓ, set

$$E_\ell(W) := W_{<\ell} W_0 W_{>\ell}.$$

If $g \in G_d^{W_0}$, then

$$(g \cdot W)_{<\ell} = g_{\ell-1} W_{<\ell} g_0^{-1}, \qquad (g \cdot W)_{>\ell} = g_L W_{>\ell} g_\ell^{-1},$$

so

$$E_\ell(g \cdot W) = g_{\ell-1} W_{<\ell} (g_0^{-1} W_0 g_L) W_{>\ell} g_\ell^{-1}.$$

Since $g_0^{-1} W_0 g_L = W_0$, we obtain

$$E_\ell(g \cdot W) = g_{\ell-1} E_\ell(W) g_\ell^{-1}.$$

Thus the vanishing of all cyclic equations is preserved. Rank is also preserved because $\mu(g \cdot W) = g_L \mu(W) g_0^{-1}$. □

Remark 3.4 (Why hidden gauge is too small)**.** Because G_{hid} preserves $\mu(W)$ exactly, it cannot move a general point of $\Gamma_{d,S}^r$ into the critical locus unless the condition

$$\mu(W) = P_S W^*$$

already holds. This is why the orbit problem must be stated with a group that acts nontrivially on the endpoints.

For positive rank, the critical locus is not stable under the full group $G_d^{W_0}$.

Proposition 3.5 (Non-stability of $\mathcal{C}_{d,S}^r$ for $r > 0$)**.** Assume $r > 0$. Then $\mathcal{C}_{d,S}^r$ is not stable under the action of $G_d^{W_0}$.

Proof. Choose $\lambda \in \mathbb{R}^\times$ with $\lambda \neq 1$, and define

$$g_0 = I_{d_0}, \qquad g_L = \lambda P_S + P_Q, \qquad g_\ell = I_{d_\ell} \ \text{ for } 1 \leq \ell \leq L-1.$$

Then g_L is invertible with inverse $\lambda^{-1} P_S + P_Q$, and

$$g_0 W_0 g_L^{-1} = \Sigma_{XY} P_Q (\lambda^{-1} P_S + P_Q) = \Sigma_{XY} P_Q = W_0.$$

So $g \in G_d^{W_0}$. If $W \in \mathcal{C}_{d,S}^r$, then

$$\mu(g \cdot W) = g_L (P_S W^*) = \lambda P_S W^*.$$

Since $r > 0$, the matrix $P_S W^*$ is nonzero, so

$$\mu(g \cdot W) \neq P_S W^*.$$

Hence $g \cdot W \notin \mathcal{C}_{d,S}^r$. □

Definition 3.6 (Correct stabilizer on the critical locus)**.** Define

$$H_S := \mathrm{Stab}_{G_d^{W_0}}(P_S W^*) = \left\{ g \in G_d^{W_0} \;\middle|\; g_L (P_S W^*) g_0^{-1} = P_S W^* \right\}.$$

Remark 3.7 (Correct orbit spaces)**.** For $r > 0$, the relevant orbit spaces are

$$\Gamma_{d,S}^r / G_d^{W_0} \qquad \text{and} \qquad \mathcal{C}_{d,S}^r / H_S.$$

The first direct weight-space theorem is precisely the statement that these two orbit spaces are in bijection.

3.2 Endpoint reduction

Definition 3.8 (Endpoint group)**.** Define the endpoint group

$$K_S := \{(g_0, g_L) \in \mathrm{GL}(d_0, \mathbb{R}) \times \mathrm{GL}(d_L, \mathbb{R}) : g_0 W_0 g_L^{-1} = W_0\}.$$

It acts on the function space $\mathrm{Mat}_{d_L \times d_0}(\mathbb{R})$ by

$$(g_0, g_L) \cdot M := g_L M g_0^{-1}.$$

Lemma 3.9 (Projection to the endpoint group)**.** The map

$$\pi : G_d^{W_0} \to K_S, \qquad \pi(g_0, \ldots, g_L) := (g_0, g_L),$$

is a surjective homomorphism, and for every $W \in \Omega$ one has

$$\mu(G_d^{W_0} \cdot W) = K_S \cdot \mu(W).$$

Proof. The projection is well-defined because $g_0 W_0 g_L^{-1} = W_0$ is exactly the defining relation of K_S. It is surjective because for every $(g_0, g_L) \in K_S$, the element

$$\widetilde{g} = (g_0, I_{d_1}, \ldots, I_{d_{L-1}}, g_L)$$

lies in $G_d^{W_0}$ and projects to (g_0, g_L). The orbit identity follows at once from equivariance of μ. □

Corollary 3.10 (Endpoint reduction)**.** For $W \in \Gamma_{d,S}^r$, the following are equivalent:

1. there exists $g \in G_d^{W_0}$ such that $g \cdot W \in \mathcal{C}_{d,S}^r$;
2. there exists $(g_0, g_L) \in K_S$ such that
$$g_L \mu(W) g_0^{-1} = P_S W^*;$$
3. the product $\mu(W)$ lies in the K_S-orbit of $P_S W^*$.

Proof. The implication (1)⇒(2) is just equivariance of μ. The implication (2)⇒(1) follows by lifting (g_0, g_L) to an element of $G_d^{W_0}$ and using stability of $\Gamma_{d,S}^r$. The equivalence between (2) and (3) is the statement that two points lie in the same orbit if and only if one is obtained from the other by the group action. □

3.3 Visible invariants and normalization

Proposition 3.11 (Visible product-level constraints)**.** If $W \in \Gamma_{d,S}$ and $M = \mu(W)$, then

$$W_0 M = 0 \qquad \text{and} \qquad M W_0 = 0.$$

Consequently,

$$\mathrm{im}(M) \subseteq \mathrm{Ker}(W_0) \qquad \text{and} \qquad \mathrm{im}(W_0) \subseteq \mathrm{Ker}(M).$$

Proof. For $\ell = 1$, the cyclic equation reads

$$W_0 W_{>1} = 0.$$

Multiplying on the right by W_1 gives

$$W_0 M = 0.$$

For $\ell = L$, the cyclic equation reads

$$W_{<L} W_0 = 0.$$

Multiplying on the left by W_L gives

$$M W_0 = 0.$$

The image-kernel statements are immediate. □

Corollary 3.12 (Rank-r sharpening)**.** Assume that Σ_{XY} has full rank d_L. If $W \in \Gamma^r_{d,S}$ and $M = \mu(W)$, then

$$\mathrm{Ker}(W_0) = \mathrm{im}(P_S), \qquad \mathrm{rank}(W_0) = d_L - r, \qquad \mathrm{im}(M) = \mathrm{im}(P_S).$$

Proof. Because Σ_{XY} has full rank d_L, it is injective as a map from $\mathbb{R}^{d_L}$ to $\mathbb{R}^{d_0}$. Hence

$$\mathrm{Ker}(W_0) = \mathrm{Ker}(\Sigma_{XY} P_Q) = \mathrm{Ker}(P_Q) = \mathrm{im}(P_S).$$

Since M has rank r and $\mathrm{im}(M) \subseteq \mathrm{Ker}(W_0)$, the inclusion is an equality. □

Theorem 3.13 (Normalization theorem for the end-to-end map)**.** Assume that Σ_{XY} has full rank d_L, and that the distinguished critical product $P_S W^*$ occurs for the subset S. Then for every

$$W \in \Gamma^r_{d,S},$$

one has

$$\mu(W) \in K_S \cdot (P_S W^*).$$

Proof. Set

$$\sigma := W_0 = \Sigma_{XY} P_Q, \qquad T := P_S W^*, \qquad U := \mathrm{Ker}(\sigma), \qquad I := \mathrm{im}(\sigma).$$

For any matrix M satisfying

$$\sigma M = 0 \qquad \text{and} \qquad M \sigma = 0,$$

the map M factors through a reduced map

$$\overline{M} : \mathbb{R}^{d_0}/I \to U.$$

Moreover, the group K_S acts on reduced maps through the natural left-right action of

$$\mathrm{GL}(U) \times \mathrm{GL}(\mathbb{R}^{d_0}/I),$$

and this action is surjective because one can choose complements of U and I and build endpoint transformations blockwise.

Now let $M = \mu(W)$ with $W \in \Gamma^r_{d,S}$. The previous proposition gives

$$\sigma M = 0 \qquad \text{and} \qquad M\sigma = 0,$$

and the rank-r sharpening shows that $\overline{M}$ is surjective. The model critical product $T = P_S W^*$ satisfies the same two annihilation relations, has rank r, and has image U. Therefore its reduced map

$$\overline{T} : \mathbb{R}^{d_0}/I \to U$$

is also surjective.

Since any two surjective linear maps from $\mathbb{R}^{d_0}/I$ to U are equivalent under left-right multiplication by invertible maps, there exists $(g_0, g_L) \in K_S$ with

$$g_L T g_0^{-1} = M.$$

Hence

$$M \in K_S \cdot T = K_S \cdot (P_S W^*),$$

as required. □

3.4 Orbit correspondence

Theorem 3.14 (First direct weight-space theorem: orbit-surjectivity)**.** Assume that Σ_{XY} has full rank d_L. Then every $G_d^{W_0}$-orbit in $\Gamma^r_{d,S}$ meets the critical locus $\mathcal{C}^r_{d,S}$.

Proof. Let $W \in \Gamma^r_{d,S}$. By Theorem 3.13,

$$\mu(W) \in K_S \cdot (P_S W^*).$$

By endpoint reduction, there exists $g \in G_d^{W_0}$ such that

$$g \cdot W \in \mathcal{C}^r_{d,S}.$$

□

Theorem 3.15 (Second direct weight-space theorem: orbit-bijection)**.** Assume that $r > 0$ and that Σ_{XY} has full rank d_L. Then every $G_d^{W_0}$-orbit in $\Gamma^r_{d,S}$ meets $\mathcal{C}^r_{d,S}$ in exactly one H_S-orbit. Equivalently, the inclusion induces a bijection

$$\mathcal{C}^r_{d,S}/H_S \xrightarrow{\sim} \Gamma^r_{d,S}/G_d^{W_0}.$$

Proof. Existence is Theorem 3.14. For uniqueness, let $W, W' \in \mathcal{C}^r_{d,S}$ lie in the same $G_d^{W_0}$-orbit. Then $W' = g \cdot W$ for some $g \in G_d^{W_0}$. Since both points lie in the critical locus,

$$\mu(W) = P_S W^* \qquad \text{and} \qquad \mu(W') = P_S W^*.$$

Equivariance gives

$$P_S W^* = \mu(W') = g_L \mu(W) g_0^{-1} = g_L (P_S W^*) g_0^{-1}.$$

Hence $g \in H_S$. This proves that

$$\left(G_d^{W_0} \cdot W\right) \cap \mathcal{C}_{d,S}^r = H_S \cdot W.$$

The bijection of orbit sets follows formally from existence and uniqueness. □

Corollary 3.16 (Transfer of stratifications)**.** Suppose that the rank-r cyclic locus admits a decomposition into locally closed $G_d^{W_0}$-stable strata

$$\Gamma_{d,S}^r = \bigsqcup_{\alpha \in A} \mathcal{O}_\alpha,$$

where each stratum is a union of $G_d^{W_0}$-orbits. Then the critical locus inherits the decomposition

$$\mathcal{C}_{d,S}^r = \bigsqcup_{\alpha \in A} \mathcal{C}_\alpha, \qquad \mathcal{C}_\alpha := \mathcal{C}_{d,S}^r \cap \mathcal{O}_\alpha,$$

and this is a stratification by locally closed H_S-stable pieces.

Proof. By Theorem 3.15, every ambient $G_d^{W_0}$-orbit meets the critical locus in exactly one H_S-orbit. Therefore the orbit partition of the cyclic locus induces an H_S-orbit partition of the critical locus. Grouping those critical orbits according to the ambient strata $\mathcal{O}_\alpha$ gives the claimed decomposition. □

Remark 3.17 (Role of the first theorem in the paper)**.** The theorem above is the direct weight-space replacement of the old lifting statement. It does not identify the cyclic locus with the critical locus. Instead, it identifies their orbit spaces in the precise way needed to transfer combinatorial data, such as cyclic rank patterns, from $\Gamma_{d,S}^r$ to $\mathcal{C}_{d,S}^r$.

Correction to the weight-space orbit correspondence. The weight-space orbit correspondence is valid in the following form.

Proposition 3.18 (Corrected normalization lemma)**.** Fix $S \subseteq \{1, \ldots, r_{\max}\}$ with $|S| = r$, and set

$$W_0 := \Sigma_{XY} P_Q, \qquad T := P_S W^*.$$

Assume Σ_{XY} has full rank d_L, and let $W \in \Gamma_{d,S}^r$. Write

$$M := \mu(W), \qquad U := \ker(W_0) \subseteq \mathbb{R}^{d_L}, \qquad I := \operatorname{im}(W_0) \subseteq \mathbb{R}^{d_0}.$$

Then:

(i) $W_0 M = 0$ and $M W_0 = 0$.

(ii) $\operatorname{im}(M) = U$ and $I \subseteq \ker(M)$.

(iii) Hence M induces a surjective linear map

$$\bar{M} : \mathbb{R}^{d_0}/I \longrightarrow U.$$

(iv) The model critical product $T = P_S W^*$ satisfies the same annihilation relations $W_0 T = 0$ and $T W_0 = 0$, has image U, and therefore induces a surjective map

$$\bar{T} : \mathbb{R}^{d_0}/I \longrightarrow U.$$

(v) If

$$K_S := \{(g_0, g_L) \in GL(d_0, \mathbb{R}) \times GL(d_L, \mathbb{R}) \mid g_0 W_0 g_L^{-1} = W_0\},$$

then the induced action of K_S on reduced maps is the natural left-right action of $GL(U) \times GL(\mathbb{R}^{d_0}/I)$. In particular, since $\bar{M}$ and $\bar{T}$ are both surjective, there exists $(g_0, g_L) \in K_S$ such that

$$g_L \, T \, g_0^{-1} = M.$$

Equivalently,

$$\mu(W) \in K_S \cdot (P_S W^*).$$

Proof. Parts (i) and (ii) are the content of Proposition 3.11 and Corollary 3.12 in the draft. The only correction needed in the original proof of Theorem 3.13 is that one must not say "M is surjective": what is surjective is the induced quotient map $\bar{M} : \mathbb{R}^{d_0}/I \to U$, because $\mathrm{im}(M) = U$. The same applies to $T = P_S W^*$. Since any two surjective linear maps from $\mathbb{R}^{d_0}/I$ onto U are equivalent under the left-right action of $GL(U) \times GL(\mathbb{R}^{d_0}/I)$, the conclusion follows. □

Theorem 3.19 (Weight-space orbit correspondence)**.** For every $0 \le r \le d_L$, define

$$G_d^{W_0} := \{g = (g_0, \ldots, g_L) \in G_d \mid g_0 W_0 g_L^{-1} = W_0\},$$

and

$$H_S := \mathrm{Stab}_{G_d^{W_0}}(P_S W^*) = \{g \in G_d^{W_0} \mid g_L (P_S W^*) g_0^{-1} = P_S W^*\}.$$

Then every $G_d^{W_0}$-orbit in $\Gamma_{d,S}^r$ meets $C_{d,S}^r$ in exactly one H_S-orbit. Equivalently, the inclusion $C_{d,S}^r \hookrightarrow \Gamma_{d,S}^r$ induces a bijection of orbit sets

$$C_{d,S}^r / H_S \xrightarrow{\sim} \Gamma_{d,S}^r / G_d^{W_0}.$$

Proof. Let $W \in \Gamma_{d,S}^r$. By the proposition above, $\mu(W) \in K_S \cdot (P_S W^*)$. By surjectivity of the endpoint projection $G_d^{W_0} \to K_S$, there exists $g \in G_d^{W_0}$ such that

$$\mu(g \cdot W) = P_S W^*.$$

Since $\Gamma_{d,S}^r$ is $G_d^{W_0}$-stable, this shows that $g \cdot W \in C_{d,S}^r$. Hence every $G_d^{W_0}$-orbit meets $C_{d,S}^r$.

For uniqueness, let $W, W' \in C_{d,S}^r$ and suppose $W' = g \cdot W$ for some $g \in G_d^{W_0}$. Then

$$P_S W^* = \mu(W') = g_L \mu(W) g_0^{-1} = g_L (P_S W^*) g_0^{-1},$$

so $g \in H_S$. Therefore

$$(G_d^{W_0} \cdot W) \cap C_{d,S}^r = H_S \cdot W,$$

which is the claimed uniqueness. □

Corollary 3.20 (What actually transfers to the critical locus)**.** Suppose

$$\Gamma^r_{d,S} = \bigsqcup_{\alpha \in A} O_\alpha$$

is a decomposition into locally closed $G_d^{W_0}$-stable subsets, each O_α being a union of $G_d^{W_0}$-orbits. Then, after discarding empty pieces,

$$C^r_{d,S} = \bigsqcup_{\alpha\,:\,C_\alpha \neq \varnothing} C_\alpha, \qquad C_\alpha := C^r_{d,S} \cap O_\alpha,$$

is a decomposition into locally closed H_S-stable subsets.

Remark 3.21 (Scope of the transfer)**.** The theorem transfers *only* decompositions of $\Gamma^r_{d,S}$ by $G_d^{W_0}$-stable unions of $G_d^{W_0}$-orbits. It does *not* by itself transfer a finer stratification coming from a smaller fixed-closing-map slice or from a hidden-gauge slice. In particular, if later sections classify a different orbit problem (for example G_σ-orbits in a fixed-arrow cyclic slice), an additional argument is required to deduce a stratification of $C^r_{d,S}$ by those finer labels.

Remark 3.22 (Full-rank case $r = d_L$)**.** If $r = d_L$, then $P_Q = 0$, hence $W_0 = 0$, so

$$G_d^{W_0} = G_d, \qquad \Gamma_{d,S} = \Omega, \qquad \Gamma^{d_L}_{d,S} = \Sigma^{d_L}_d.$$

The theorem becomes

$$C^{d_L}_{d,S}/H_S \xrightarrow{\sim} \Sigma^{d_L}_d/G_d.$$

Thus the result is compatible with the explicit normal form of all global minimizers, but it is strictly coarser than a stratification of the fixed fiber $\mu^{-1}(P_S W^*)$: it identifies orbit quotients, not the finer geometry of the fiber itself.

Remark 3.23 (Depth $L = 3$)**.** For $L = 3$, the proof is unchanged. The only cyclic relations used in the normalization argument are the endpoint relations implying

$$W_0 \mu(W) = 0, \qquad \mu(W) W_0 = 0,$$

and the rest is linear algebra on the end-to-end product. Hence no extra degeneracy appears at depth 3.

Remark 3.24 (Full-rank case and relation with Simon's orbit correspondence)**.** The orbit-bijection theorem is compatible with the full-rank case $r = d_L$. Indeed, if $r = d_L$, then $P_Q = 0$, hence

$$W_0 = \Sigma_{XY} P_Q = 0, \qquad G_d^{W_0} = G_d, \qquad \Gamma^{d_L}_{d,S} = \Sigma^{d_L}_d.$$

Therefore Theorem 3.15 specializes to

$$C^{d_L}_{d,S}/H_S \xrightarrow{\sim} \Sigma^{d_L}_d/G_d.$$

Equivalently, every G_d-orbit in the rank-d_L locus $\Sigma^{d_L}_d$ meets the fixed-product fiber $C^{d_L}_{d,S}$ in exactly one H_S-orbit.

Thus, in the full-rank case, the theorem recovers the same orbit-correspondence principle as in Simon's work on the fiber of the multiplication map. In particular, if one chooses the decomposition of $\Sigma_d^{d_L}$ by individual G_d-orbits,

$$\Sigma_d^{d_L} = \bigsqcup_{\lambda} O_\lambda,$$

then the induced decomposition

$$C_{d,S}^{d_L} = \bigsqcup_{\lambda:\, O_\lambda \cap C_{d,S}^{d_L} \neq \varnothing} \left(O_\lambda \cap C_{d,S}^{d_L}\right)$$

is exactly the corresponding orbit decomposition of the fiber.

The only sense in which a transferred decomposition may be coarser is the following: Corollary 3.16 transfers whatever $G_d^{W_0}$-stable decomposition of $\Gamma_{d,S}^r$ one starts with. If one uses a decomposition by unions of $G_d^{W_0}$-orbits rather than by individual orbits, then the induced decomposition of $C_{d,S}^r$ is correspondingly coarser. This is a feature of the chosen stratification, not a defect of Theorem 3.15 itself.

4 Stratification of Cyclic Quivers with Finite Relation

This section has two layers. First, we work in the ambient bound-quiver setting and classify all representations of the algebra

$$A_L^{\mathrm{wt}} = kQ/I_L^{\mathrm{wt}}$$

for a free closing arrow a_0. This is the intrinsic representation-theoretic classification. Second, we specialize to the deep linear network by freezing the closing arrow to

$$W_0 = \Sigma_{XY} P_Q,$$

thereby passing to the fixed-arrow slice that defines $\Gamma_{d,S}$. The second conjecture belongs to this second, fixed-arrow level.

Throughout this section we assume

$$L \geq 2,$$

which is the first genuinely cyclic case. For the purely algebraic statements we work over a field k; when we return to the deep linear network we specialize to $k = \mathbb{R}$ and to the distinguished closing map

$$W_0 = \Sigma_{XY} P_Q.$$

4.1 The cyclic quiver and the fixed-arrow slice

Definition 4.1 (The cyclic quiver and the weight-space ideal)**.** Let

$$Q = C_{L+1}$$

be the oriented cycle on vertices

$$Q_0 = \{0, 1, \ldots, L\}$$

with arrows

$$a_t : t-1 \to t \qquad (1 \le t \le L), \qquad a_0 : L \to 0.$$

All arrow indices are read modulo $L+1$.

For each $\ell \in \{1, \dots, L\}$, define the length-L path

$$p_\ell := a_{\ell-1} a_{\ell-2} \cdots a_{\ell-L} = a_{\ell-1} \cdots a_1\, a_0\, a_L \cdots a_{\ell+1}.$$

This is the unique oriented path of length L that omits the arrow a_ℓ. Let

$$p_0 := a_L \cdots a_1, \qquad I_L^{\mathrm{wt}} := \langle p_1, \dots, p_L \rangle \subseteq kQ, \qquad A_L^{\mathrm{wt}} := kQ/I_L^{\mathrm{wt}}.$$

Remark 4.2 (The surviving open-chain path)**.** Every generator p_ℓ, $1 \le \ell \le L$, passes through the closing arrow a_0. The only oriented path of length L that does not pass through a_0 is

$$p_0 = a_L \cdots a_1.$$

Under the weight-space dictionary this path acts as the end-to-end product $\mu(W) = W_L \cdots W_1$, so it must survive in the direct weight-space formalism.

Definition 4.3 (Representation varieties)**.** Fix a dimension vector

$$d = (d_0, \dots, d_L)$$

and vector spaces

$$V_\ell \cong k^{d_\ell}, \qquad 0 \le \ell \le L.$$

Set

$$\mathrm{Rep}_d(Q) := \prod_{\ell=1}^{L} \mathrm{Hom}(V_{\ell-1}, V_\ell) \times \mathrm{Hom}(V_L, V_0),$$

and let

$$\mathrm{Rep}_d(Q, I_L^{\mathrm{wt}}) := \{\phi \in \mathrm{Rep}_d(Q) \mid p_\ell(\phi) = 0 \text{ for all } 1 \le \ell \le L\}.$$

We also write

$$\mathrm{Rep}_d(A_L^{\mathrm{wt}}) := \mathrm{Rep}_d(Q, I_L^{\mathrm{wt}})$$

for the affine representation variety of the bound-quiver algebra A_L^{wt} with dimension vector d. For a fixed map

$$W_0 \in \mathrm{Hom}(V_L, V_0),$$

define the fixed-arrow slice

$$\mathrm{Rep}_{d,W_0}(Q, I_L^{\mathrm{wt}}) := \{\phi \in \mathrm{Rep}_d(Q, I_L^{\mathrm{wt}}) \mid \phi_0 = W_0\}.$$

Proposition 4.4 (Bound-quiver interpretation)**.** Let

$$A_L^{\mathrm{wt}} := kQ/I_L^{\mathrm{wt}}.$$

The standard functor

$$\mathrm{Rep}_k(Q, I_L^{\mathrm{wt}}) \longrightarrow A_L^{\mathrm{wt}}\text{-mod}, \qquad (V_i, \phi_a) \longmapsto \bigoplus_{i=0}^{L} V_i,$$

where each primitive idempotent e_i acts as the projector onto V_i and each arrow a acts as ϕ_a, is an equivalence of categories. For fixed spaces $V_i \cong k^{d_i}$ and a fixed map

$$W_0 \in \mathrm{Hom}(V_L, V_0),$$

the affine variety

$$\mathrm{Rep}_{d,W_0}(Q, I_L^{\mathrm{wt}})$$

is exactly the variety of those objects of $\mathrm{Rep}_k(Q, I_L^{\mathrm{wt}})$ having dimension vector d and whose a_0-map is W_0. Thus the fixed-arrow slice is a representation variety inside $\mathrm{Rep}_k(Q, I_L^{\mathrm{wt}})$, not a separate module category.

Proof. The first statement is the standard equivalence between finite-dimensional representations of a bound quiver and finite-dimensional modules over its bound-quiver algebra. The second statement is simply the definition of the fiber cut out by the affine linear equation

$$\phi_0 = W_0$$

inside the representation variety $\mathrm{Rep}_d(Q, I_L^{\mathrm{wt}})$. □

Proposition 4.5 (The cyclic equations are the quiver relations)**.** Specialize to $k = \mathbb{R}$ and fix the closing map

$$W_0 = \Sigma_{XY} P_Q.$$

For a weight tuple

$$W = (W_1, \ldots, W_L) \in \Omega,$$

define

$$\phi(W) := (W_0, W_1, \ldots, W_L) \in \mathrm{Rep}_d(Q).$$

Then for every $1 \le \ell \le L$,

$$p_\ell(\phi(W)) = \phi_{\ell-1}\phi_{\ell-2}\cdots\phi_{\ell-L} = W_{<\ell} W_0 W_{>\ell}.$$

Consequently,

$$W \in \Gamma_{d,S} \quad \Longleftrightarrow \quad \phi(W) \in \mathrm{Rep}_{d,W_0}(Q, I_L^{\mathrm{wt}}).$$

Proof. By definition,

$$p_\ell = a_{\ell-1}a_{\ell-2}\cdots a_{\ell-L} = a_{\ell-1}\cdots a_1\, a_0\, a_L \cdots a_{\ell+1}.$$

Substituting

$$\phi_0 = W_0, \qquad \phi_t = W_t \quad (1 \le t \le L),$$

gives

$$p_\ell(\phi(W)) = W_{\ell-1}\cdots W_1\, W_0\, W_L \cdots W_{\ell+1} = W_{<\ell} W_0 W_{>\ell}.$$

The final equivalence is therefore exactly the definition of the cyclic locus. □

Corollary 4.6 (The cyclic locus as a fixed-arrow representation variety)**.** The map

$$W \longmapsto \phi(W)$$

identifies the weight-space cyclic locus $\Gamma_{d,S}$ with the fixed-arrow slice $\mathrm{Rep}_{d,W_0}(Q, I_L^{\mathrm{wt}})$.

4.2 The weight-space cyclic algebra

Lemma 4.7 (Surviving and vanishing paths)**.** In the quotient algebra A_L^{wt}:

1. the path
$$p_0 = a_L \cdots a_1$$
survives;
2. every other oriented path of length L is zero;
3. every oriented path of length greater than L is zero.

Proof. The generators of I_L^{wt} are exactly the length-L paths
$$p_1, \ldots, p_L,$$
namely the oriented length-L paths that pass through the closing arrow a_0. The remaining length-L path is
$$p_0 = a_L \cdots a_1,$$
which does not pass through a_0, so its class is not killed by this monomial ideal.

Any oriented path of length $L+1$ goes once around the entire cycle, hence contains a length-L subpath passing through a_0, and therefore contains one of the generators $p_1, \ldots, p_L$. It follows that every path of length $L+1$ lies in I_L^{wt}. Since I_L^{wt} is a two-sided ideal, the same is true for every path of length greater than $L+1$. □

Corollary 4.8 (Projective lengths)**.** Let
$$P(i) := A_L^{\mathrm{wt}} e_i$$
be the indecomposable projective at vertex i. Then
$$\ell(P(0)) = L+1, \qquad \ell(P(i)) = L \quad (1 \le i \le L).$$
Equivalently, the Kupisch series of A_L^{wt} is
$$(L+1, L, \ldots, L).$$

Proof. Starting at vertex 0, the maximal nonzero oriented path has length L, namely p_0, so $P(0)$ has composition length $L+1$.

If $1 \le i \le L$, then the unique length-L path starting at i is p_i, which vanishes in the quotient. Every shorter path survives, so the maximal nonzero path starting at i has length $L-1$. Hence $P(i)$ has composition length L. □

Theorem 4.9 (The weight-space algebra is Nakayama)**.** The algebra
$$A_L^{\mathrm{wt}} = kQ/I_L^{\mathrm{wt}}$$
is a cyclic monomial Nakayama algebra with Kupisch series
$$(L+1, L, \ldots, L).$$

Proof. By Lemma 4.7, all paths of length greater than L vanish, so A_L^{wt} is finite-dimensional. The ideal I_L^{wt} is generated by paths, hence the algebra is monomial.

The quiver $Q = C_{L+1}$ is connected and each vertex has exactly one incoming arrow and one outgoing arrow. Since

$$(kQ)_{\geq L+1} \subseteq I_L^{\mathrm{wt}} \subseteq (kQ)_{\geq 2},$$

the ideal is admissible, and the standard bound-quiver criterion shows that the quotient is a cyclic Nakayama algebra. The Kupisch series was computed in Corollary 4.8. □

4.3 Indecomposable interval modules

Definition 4.10 (Oriented intervals and admissible indices)**.** For vertices $i, j \in \{0, \ldots, L\}$, let

$$[i \to j]_0$$

denote the oriented interval obtained by starting at i, following the orientation of the cycle, and stopping at j, with no repeated vertices. Let

$$[i \to j]_1$$

be the set of arrows occurring in this path, and let

$$\delta(i, j) := \#[i \to j]_1$$

be its oriented length. The admissible index set is

$$\mathcal{I}_L^{\mathrm{wt}} := \{(i, j) \mid 0 \leq i \leq j \leq L\} \cup \{(i, j) \mid 1 \leq i \leq L,\ 0 \leq j \leq i - 2\}.$$

Definition 4.11 (Interval modules)**.** For a pair (i, j) with $0 \leq \delta(i, j) \leq L$, define the interval representation $M(i, j)$ by

$$M(i, j)_t = \begin{cases} k, & t \in [i \to j]_0, \\ 0, & t \notin [i \to j]_0, \end{cases}$$

and for each arrow $a_s : s - 1 \to s$,

$$M(i, j)_{a_s} = \begin{cases} \mathrm{id}_k, & a_s \in [i \to j]_1, \\ 0, & a_s \notin [i \to j]_1. \end{cases}$$

Proposition 4.12 (Admissibility criterion)**.** The interval representation $M(i, j)$ factors through the quotient algebra A_L^{wt} if and only if

$$(i, j) \in \mathcal{I}_L^{\mathrm{wt}}.$$

Proof. If

$$0 \leq i \leq j \leq L,$$

then the support interval lies on the open chain

$$0 \to 1 \to \cdots \to L,$$

so the closing arrow a_0 acts by zero. Every relation path p_ℓ passes through a_0, hence acts by zero as well.

If

$$1 \le i \le L, \qquad 0 \le j \le i-2,$$

then $[i \to j]$ is a wrapped interval whose length is

$$\delta(i,j) = L + j - i + 1 \le L - 1.$$

No relation path p_ℓ of length L can be contained in its arrow support, so again every p_ℓ acts by zero.

Finally, if $1 \le i \le L$, the forbidden full wrapped interval $[i \to i-1]$ has arrow support exactly equal to the relation path p_i. Hence p_i acts as the identity on the one-dimensional space at vertex i, so $M(i, i-1)$ does not factor through A_L^{wt}. □

Proposition 4.13 (Projective-quotient realization)**.** For every admissible pair

$$(i,j) \in \mathcal{I}_L^{\mathrm{wt}},$$

there is an isomorphism

$$M(i,j) \cong P(i)/\operatorname{rad}^{\delta(i,j)+1} P(i).$$

Proof. By Theorem 4.9, each indecomposable projective $P(i)$ is uniserial, with composition length

$$c_i = \begin{cases} L+1, & i = 0, \\ L, & 1 \le i \le L. \end{cases}$$

The admissibility condition is equivalent to

$$\delta(i,j) \le c_i - 1,$$

so there is a unique indecomposable quotient of $P(i)$ of length $\delta(i,j)+1$. Its support is precisely the first $\delta(i,j)+1$ simples encountered along the oriented cycle starting at i, which is exactly the interval module $M(i,j)$. □

Theorem 4.14 (Indecomposables of the cyclic Nakayama algebra)**.** The finite-dimensional indecomposable A_L^{wt}-modules are precisely the interval modules

$$M(i,j) \qquad ((i,j) \in \mathcal{I}_L^{\mathrm{wt}}),$$

and each indecomposable occurs uniquely in this form.

Proof. Since A_L^{wt} is Nakayama, every indecomposable module is uniserial and is a quotient of a unique indecomposable projective $P(i)$. The admissible quotients of $P(i)$ are exactly the modules

$$P(i)/\operatorname{rad}^m P(i) \qquad 1 \le m \le \ell(P(i)),$$

and by Proposition 4.13 these are exactly the interval modules $M(i,j)$ with

$$(i,j) \in \mathcal{I}_L^{\mathrm{wt}}.$$

Uniqueness follows because the top simple determines the start vertex i, and the composition length determines the endpoint j along the oriented interval starting at i. □

4.4 Free-arrow and fixed-arrow orbit classification

Definition 4.15 (Multiplicity-pattern set)**.** For an admissible pair $(i,j) \in \mathcal{I}_L^{\mathrm{wt}}$ and a vertex $k \in \{0,\dots,L\}$, define

$$\mathbf{1}_{[i\to j]}(k) := \begin{cases} 1, & k \in [i \to j]_0, \\ 0, & k \notin [i \to j]_0. \end{cases}$$

Then set

$$\mathcal{M}_d^+(A_L^{\mathrm{wt}}) := \left\{ (m(i,j))_{(i,j)\in\mathcal{I}_L^{\mathrm{wt}}} \in \mathbb{Z}_{\geq 0}^{\mathcal{I}_L^{\mathrm{wt}}} \;\middle|\; d_k = \sum_{(i,j)\in\mathcal{I}_L^{\mathrm{wt}}} m(i,j)\,\mathbf{1}_{[i\to j]}(k) \text{ for all } k \right\}.$$

Proposition 4.16 (Krull–Schmidt decomposition)**.** Every finite-dimensional A_L^{wt}-module of dimension vector d decomposes uniquely as

$$M \cong \bigoplus_{(i,j)\in\mathcal{I}_L^{\mathrm{wt}}} M(i,j)^{\oplus m(i,j)}$$

for a unique multiplicity pattern

$$m = (m(i,j)) \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}).$$

Proof. The algebra A_L^{wt} is finite-dimensional, hence artinian, so its module category is Krull–Schmidt. By Theorem 4.14, the indecomposable summands are exactly the admissible interval modules $M(i,j)$. The condition that the direct sum have dimension vector d is precisely the displayed formula defining $\mathcal{M}_d^+(A_L^{\mathrm{wt}})$. □

Proposition 4.17 (Ambient free-arrow orbit classification)**.** Let

$$G_d := \prod_{\ell=0}^{L} \mathrm{GL}(d_\ell, k)$$

act on $\mathrm{Rep}_d(Q)$ by base change,

$$(g\cdot\phi)_\ell := g_\ell\,\phi_\ell\,g_{\ell-1}^{-1} \qquad (0 \leq \ell \leq L).$$

Then the assignment sending a representation to the multiplicities of its indecomposable interval summands induces a canonical bijection

$$G_d \backslash \mathrm{Rep}_d(A_L^{\mathrm{wt}}) \xrightarrow{\sim} \mathcal{M}_d^+(A_L^{\mathrm{wt}}).$$

Proof. Two points of $\mathrm{Rep}_d(A_L^{\mathrm{wt}})$ lie in the same G_d-orbit if and only if they are isomorphic as quiver representations, equivalently as A_L^{wt}-modules. By Proposition 4.16, isomorphism classes of modules of dimension vector d are uniquely determined by their multiplicity patterns in $\mathcal{M}_d^+(A_L^{\mathrm{wt}})$. □

Remark 4.18 (Free arrow versus fixed arrow)**.** The indecomposable A_L^{wt}-modules do not change when the closing arrow is specialized. The ambient free-arrow problem and the fixed-arrow DLN slice use the same indecomposable interval modules from Theorem 4.14. Fixing

$$a_0 = W_0$$

does not create a new indecomposable classification; it only restricts which multiplicity patterns occur in the slice. That specialization is detected by the scalar functional ν_0 introduced below.

Definition 4.19 (The module attached to a cyclic point)**.** For

$$W \in \Gamma_{d,S},$$

let

$$M(W)$$

denote the A_L^{wt}-module corresponding to $\phi(W)$ under Corollary 4.6. Write its Krull–Schmidt decomposition as

$$M(W) \cong \bigoplus_{(i,j)\in\mathcal{I}_L^{\mathrm{wt}}} M(i,j)^{\oplus m_W(i,j)}.$$

Proposition 4.20 (Orbit classification in the fixed-arrow slice)**.** For

$$W, W' \in \Gamma_{d,S},$$

one has

$$W' \in G_d^{W_0} \cdot W \quad \Longleftrightarrow \quad M(W') \cong M(W)$$

as A_L^{wt}-modules. Consequently, the $G_d^{W_0}$-orbits in $\Gamma_{d,S}$ are exactly the isomorphism classes of A_L^{wt}-modules that admit a representative of dimension vector d with $a_0 = W_0$.

Proof. If $W' \in G_d^{W_0} \cdot W$, then

$$\phi(W') = g \cdot \phi(W)$$

for some $g = (g_0, \dots, g_L) \in G_d^{W_0}$, so g is an isomorphism of quiver representations and hence of the corresponding A_L^{wt}-modules.

Conversely, assume

$$M(W') \cong M(W).$$

Then the corresponding quiver representations are isomorphic, so there exists

$$g = (g_0, \dots, g_L) \in G_d$$

with

$$\phi(W')_\ell = g_\ell \, \phi(W)_\ell \, g_{\ell-1}^{-1} \qquad (0 \le \ell \le L).$$

At $\ell = 0$ both representations have closing arrow W_0, hence

$$W_0 = \phi(W')_0 = g_0 \phi(W)_0 g_L^{-1} = g_0 W_0 g_L^{-1}.$$

Thus $g \in G_d^{W_0}$ and $W' \in G_d^{W_0} \cdot W$. □

Definition 4.21 (Closing-arrow rank functional)**.** For a multiplicity pattern

$$m = (m(i,j))_{(i,j)\in\mathcal{I}_L^{\mathrm{wt}}} \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}),$$

we denote the specialization rank functional by $\nu_0(m)$, to avoid a clash with the DLN rank parameter r, and define

$$\nu_0(m) := \sum_{\substack{(i,j)\in\mathcal{I}_L^{\mathrm{wt}} \\ a_0 \in [i\to j]_1}} m(i,j) = \sum_{\substack{1\le i\le L \\ 0\le j\le i-2}} m(i,j).$$

Remark 4.22. The scalar $\nu_0(m)$ is the analogue, for the present weight-space setup, of the specialization functional that detects when an ambient free-arrow orbit meets the fixed-arrow slice.

Definition 4.23 (Realized multiplicity patterns and strata)**.** Define

$$\mathcal{M}^{\mathrm{real}}_{d,S}(A^{\mathrm{wt}}_L) := \{m_W \mid W \in \Gamma_{d,S}\} \subseteq \mathcal{M}^+_d(A^{\mathrm{wt}}_L).$$

For

$$m \in \mathcal{M}^+_d(A^{\mathrm{wt}}_L),$$

define

$$\Gamma_{d,S}[m] := \{W \in \Gamma_{d,S} \mid m_W = m\}, \qquad \Gamma^r_{d,S}[m] := \Gamma_{d,S}[m] \cap \Gamma^r_{d,S}.$$

Proposition 4.24 (Explicit characterization of realized multiplicity patterns)**.** For a multiplicity pattern

$$m = (m(i,j))_{(i,j)\in\mathcal{I}^{\mathrm{wt}}_L} \in \mathcal{M}^+_d(A^{\mathrm{wt}}_L),$$

the scalar $\nu_0(m)$ is the rank of the closing-arrow map on the corresponding ambient orbit. Consequently,

$$\mathcal{M}^{\mathrm{real}}_{d,S}(A^{\mathrm{wt}}_L) = \{m \in \mathcal{M}^+_d(A^{\mathrm{wt}}_L) \mid \nu_0(m) = \mathrm{rank}(W_0)\}.$$

Equivalently, in the DLN setting,

$$\mathcal{M}^{\mathrm{real}}_{d,S}(A^{\mathrm{wt}}_L) = \{m \in \mathcal{M}^+_d(A^{\mathrm{wt}}_L) \mid \nu_0(m) = d_L - r\}.$$

Proof. On an indecomposable interval module $M(i,j)$, the arrow $a_0 : L \to 0$ acts with rank 1 if and only if the interval $[i \to j]$ contains the arrow a_0, equivalently if and only if

$$1 \le i \le L, \qquad 0 \le j \le i-2.$$

Otherwise a_0 acts by 0. Hence on the direct sum determined by m, the total rank of the a_0-map is exactly $\nu_0(m)$.

If $m = m_W$ is realized by a point $W \in \Gamma_{d,S}$, then the a_0-map is exactly W_0, so necessarily

$$\nu_0(m) = \mathrm{rank}(W_0).$$

Conversely, let

$$m \in \mathcal{M}^+_d(A^{\mathrm{wt}}_L)$$

satisfy

$$\nu_0(m) = \mathrm{rank}(W_0),$$

and let

$$M_m := \bigoplus_{(i,j)\in\mathcal{I}^{\mathrm{wt}}_L} M(i,j)^{\oplus m(i,j)}.$$

The action of a_0 on M_m is a linear map $V_L \to V_0$ of rank $\nu_0(m)$. Since two matrices are equivalent under $\mathrm{GL}(V_0) \times \mathrm{GL}(V_L)$ if and only if they have the same rank, there exist endpoint changes of basis sending the a_0-map of M_m to the prescribed matrix W_0. After transporting the remaining arrow maps by the same base changes, one obtains a representation in

$$\mathrm{Rep}_{d,W_0}(Q, I^{\mathrm{wt}}_L),$$

hence a point of $\Gamma_{d,S}$ with multiplicity pattern m.

The last displayed description follows from

$$\operatorname{rank}(W_0) = \operatorname{rank}(\Sigma_{XY} P_Q) = d_L - r$$

under the standing assumption that Σ_{XY} has full rank d_L. □

Theorem 4.25 (Second conjecture: multiplicity-pattern stratification of the fixed-arrow cyclic locus)**.** The assignment

$$W \longmapsto m_W$$

induces a bijection

$$G_d^{W_0} \backslash \Gamma_{d,S} \xrightarrow{\sim} \{m \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}) \mid \nu_0(m) = \operatorname{rank}(W_0)\}.$$

Equivalently,

$$\Gamma_{d,S} = \bigsqcup_{\substack{m \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}) \\ \nu_0(m) = \operatorname{rank}(W_0)}} \Gamma_{d,S}[m],$$

and each nonempty stratum $\Gamma_{d,S}[m]$ is a single $G_d^{W_0}$-orbit. Over an algebraically closed field these orbits are Zariski locally closed; over $k = \mathbb{R}$, they are semialgebraic and locally closed in the Euclidean topology.

Proof. By Proposition 4.20, two points of $\Gamma_{d,S}$ lie in the same $G_d^{W_0}$-orbit if and only if they define isomorphic A_L^{wt}-modules. By Krull–Schmidt, this happens if and only if they have the same multiplicity pattern. Proposition 4.24 identifies exactly which multiplicity patterns occur in the fixed-arrow slice, namely those satisfying

$$\nu_0(m) = \operatorname{rank}(W_0).$$

This proves the bijection and the orbit decomposition. □

Remark 4.26 (Full-rank case $r = d_L$)**.** If

$$r = d_L,$$

then

$$W_0 = \Sigma_{XY} P_Q = 0,$$

so

$$\nu_0(m) = 0.$$

Hence no wrapped interval can occur in a realized multiplicity pattern. The only realized indecomposables are therefore

$$M(i,j) \qquad (0 \le i \le j \le L),$$

namely the ordinary interval modules of the equioriented chain

$$0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow L.$$

Equivalently,

$$\operatorname{Rep}_{d,0}(Q, I_L^{\mathrm{wt}}) \cong \operatorname{Rep}_d(0 \to 1 \to \cdots \to L),$$

because every relation in I_L^{wt} contains the arrow a_0.

Example 4.27 (Depth $L = 3$)**.** For

$$L = 3,$$

the admissible index set is

$$\mathcal{I}_3^{\mathrm{wt}} = \{(0,0),(0,1),(0,2),(0,3),(1,1),(1,2),(1,3),(2,2),(2,3),(2,0),(3,3),(3,0),(3,1)\}.$$

Hence the indecomposable A_3^{wt}-modules are precisely

$$\begin{aligned}
&M(0,0),\ M(0,1),\ M(0,2),\ M(0,3),\\
&M(1,1),\ M(1,2),\ M(1,3),\\
&M(2,2),\ M(2,3),\ M(2,0),\\
&M(3,3),\ M(3,0),\ M(3,1).
\end{aligned}$$

Explicitly, their supports are

$$\begin{aligned}
&M(0,0) = S_0, \quad M(0,1) : 0 \to 1, \quad M(0,2) : 0 \to 1 \to 2, \quad M(0,3) : 0 \to 1 \to 2 \to 3,\\
&M(1,1) = S_1, \quad M(1,2) : 1 \to 2, \quad M(1,3) : 1 \to 2 \to 3,\\
&M(2,2) = S_2, \quad M(2,3) : 2 \to 3, \quad M(2,0) : 2 \to 3 \to 0,\\
&M(3,3) = S_3, \quad M(3,0) : 3 \to 0, \quad M(3,1) : 3 \to 0 \to 1.
\end{aligned}$$

The forbidden full wrapped intervals

$$M(1,0), \qquad M(2,1), \qquad M(3,2)$$

do not factor through A_3^{wt}, because they realize the relation paths p_1, p_2, p_3, respectively. The indecomposables containing the closing arrow a_0 are exactly

$$M(2,0), \qquad M(3,0), \qquad M(3,1),$$

so for a multiplicity pattern m one has

$$\nu_0(m) = m(2,0) + m(3,0) + m(3,1).$$

Therefore, in the DLN fixed-arrow slice,

$$m \in \mathcal{M}_{d,S}^{\mathrm{real}}(A_3^{\mathrm{wt}}) \quad \Longleftrightarrow \quad m(2,0) + m(3,0) + m(3,1) = \mathrm{rank}(W_0) = d_3 - r.$$

In particular, if

$$r = d_3,$$

then only the ten chain intervals

$$M(0,0), M(0,1), M(0,2), M(0,3), M(1,1), M(1,2), M(1,3), M(2,2), M(2,3), M(3,3)$$

are realized.

5 Application: Stratification of the Critical Loci

Corollary 5.1 (Rank-r strata and transfer to the critical locus)**.** Define

$$\mathcal{M}_{d,S}^{\mathrm{real},r}(A_L^{\mathrm{wt}}) := \{m \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}) \mid \nu_0(m) = d_L - r,\ \Gamma_{d,S}^r[m] \neq \varnothing\}.$$

Then

$$\Gamma_{d,S}^r = \bigsqcup_{m \in \mathcal{M}_{d,S}^{\mathrm{real},r}(A_L^{\mathrm{wt}})} \Gamma_{d,S}^r[m]$$

is a decomposition into $G_d^{W_0}$-orbits. By Theorem 3.15, this transfers to

$$\mathcal{C}_{d,S}^r = \bigsqcup_{m \in \mathcal{M}_{d,S}^{\mathrm{real},r}(A_L^{\mathrm{wt}})} \mathcal{C}_{d,S}^r[m], \qquad \mathcal{C}_{d,S}^r[m] := \mathcal{C}_{d,S}^r \cap \Gamma_{d,S}^r[m],$$

and each nonempty $\mathcal{C}_{d,S}^r[m]$ is a single H_S-orbit.

Proof. Theorem 4.25 shows that each nonempty

$$\Gamma_{d,S}[m]$$

is a single $G_d^{W_0}$-orbit. Intersecting with the rank-r locus simply discards the patterns that do not occur there, which yields the displayed decomposition of $\Gamma_{d,S}^r$.

Theorem 3.15 identifies each $G_d^{W_0}$-orbit in $\Gamma_{d,S}^r$ with a unique H_S-orbit in the critical locus. Applying this orbit correspondence to the decomposition above yields the claimed stratification of $\mathcal{C}_{d,S}^r$. □

6 Discussion and Future Work

The first direct weight-space theorem identifies the orbit spaces of the cyclic locus and the critical locus. The second conjecture, proved above, identifies the orbit labels on the cyclic side: they are the realized multiplicity patterns of the admissible interval modules for the cyclic Nakayama algebra A_L^{wt}.

What remains is the finer geometry inside this stratification: closure order, codimension, Ext-groups, and intrinsic criteria describing which multiplicity patterns survive additional rank conditions in the fixed-arrow slice. These are the natural next steps for understanding how the critical strata sit inside the ambient weight space and how they should interact with the saddle-combinatorics questions motivating the project.